

Envelope Theorem

- 2 endogene variable (x_1, x_2) & 1 bivæbningelse

- Værdifunktionen $M(a) = \max_{\{x_1, x_2\}} f(x_1, x_2, a)$

$$\text{s.t. } h(x_1, x_2, a) = 0$$

- Optimal Lagrange: $\mathcal{L} = f(x_1, x_2, a) - \lambda(h(x_1, x_2, a))$
- Ud fra F.O.C finder vi $(x_1^*(a), x_2^*(a))$ som indsæt i f giver os den optimale værdifkt.:

$$M(a) = f(x^*(a), x^*(a), a) \quad (1)$$

Envelope Theorem

$$\begin{aligned} \frac{dM(a)}{da} &= \frac{\partial \mathcal{L}(x_1^*(a), x_2^*(a), a)}{\partial a} \\ &= \frac{\partial f(x_1^*(a), x_2^*(a), a)}{\partial a} - \lambda \frac{\partial h(x_1^*(a), x_2^*(a), a)}{\partial a} \end{aligned}$$

Beweis:

- Diff. (1) ifht. a :

$$\frac{dM(a)}{da} = \frac{\partial f}{\partial x_1} \frac{dx_1}{da} + \frac{\partial f}{\partial x_2} \frac{dx_2}{da} + \frac{\partial f}{\partial a} \quad (2)$$

- Betragt F.O.C fra lagrange-problemet:

$$(1) \frac{\partial \mathcal{L}}{\partial x_1} = \frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0 \Leftrightarrow \frac{\partial f}{\partial x_1} = \lambda \frac{\partial h}{\partial x_1} \quad (*)$$

$$(2) \frac{\partial \mathcal{L}}{\partial x_2} = \frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0 \Leftrightarrow \frac{\partial f}{\partial x_2} = \lambda \frac{\partial h}{\partial x_2} \quad (**)$$

$$(3) \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Leftrightarrow h(x_1, x_2, a) = 0 \quad (***)$$

- Indsæt $(*)$ & $(**)$ i (2):

$$\begin{aligned} \Rightarrow \frac{dM(a)}{da} &= \lambda \frac{\partial h}{\partial x_1} \frac{dx_1}{da} + \lambda \frac{\partial h}{\partial x_2} \frac{dx_2}{da} + \frac{\partial f}{\partial a} \\ &= \lambda \left[\frac{\partial h}{\partial x_1} \frac{dx_1}{da} + \frac{\partial h}{\partial x_2} \frac{dx_2}{da} \right] + \frac{\partial f}{\partial a} \quad (3) \end{aligned}$$

- I optimum er bivæbningen bindende $(***)$ (fra F.O.C):

$$h(x_1^*(a), x_2^*(a), a) = 0$$

$$\Rightarrow \frac{d h(x_1^*(a), x_2^*(a), a)}{da} = 0$$

$$\frac{\partial h}{\partial x_1} \frac{dx_1}{da} + \frac{\partial h}{\partial x_2} \frac{dx_2}{da} + \frac{\partial h}{\partial a} = 0 \Leftrightarrow \frac{\partial h}{\partial x_1} \frac{dx_1}{da} + \frac{\partial h}{\partial x_2} \frac{dx_2}{da} = - \frac{\partial h}{\partial a} (\Delta)$$

• Indsæt (1) i (3):

$$\Rightarrow \frac{dM(a)}{da} = \lambda \left[-\frac{\partial h}{\partial a} \right] + \frac{\partial f}{\partial a} = \frac{\partial f}{\partial a} - \lambda \frac{\partial h}{\partial a} = \frac{\partial \mathcal{L}}{\partial a} \quad \blacksquare$$